

Hype vs. Value: Can Valuation-Based Scoring Predict Market Returns?

1. Research Question

Does a stock's relative valuation — measured as its current P/E ratio compared to its own 5-year historical average — predict its 1-year forward return?

The hypothesis: markets systematically misprice assets relative to their own fundamentals. If a stock trades significantly above its historical valuation baseline, mean reversion should produce lower subsequent returns — and vice versa. To test this, I built a metric called the **Hype Score** and applied it to the entire S&P 500 universe over 14 years.

2. Methodology

Data Scope

Dimension	Detail
Universe	S&P 500 — all historical constituents (not just current members)
Time Range	June 2011 – February 2026
Records	1.85 million daily observations, 635 companies
Sources	SEC EDGAR (quarterly EPS), Yahoo Finance (prices), Wikipedia (index membership)

How Fair Value Is Calculated

Each stock receives a daily Fair Value based on its own earnings history:

1. **TTM EPS** — Rolling sum of the 4 most recent quarterly EPS reports
2. **Daily P/E** — Close Price ÷ TTM EPS
3. **5-Year Average P/E** — Rolling mean of daily P/E over the prior 1,260 trading days

4. **Fair Value** — 5-Year Average P/E × TTM EPS

Hype Score

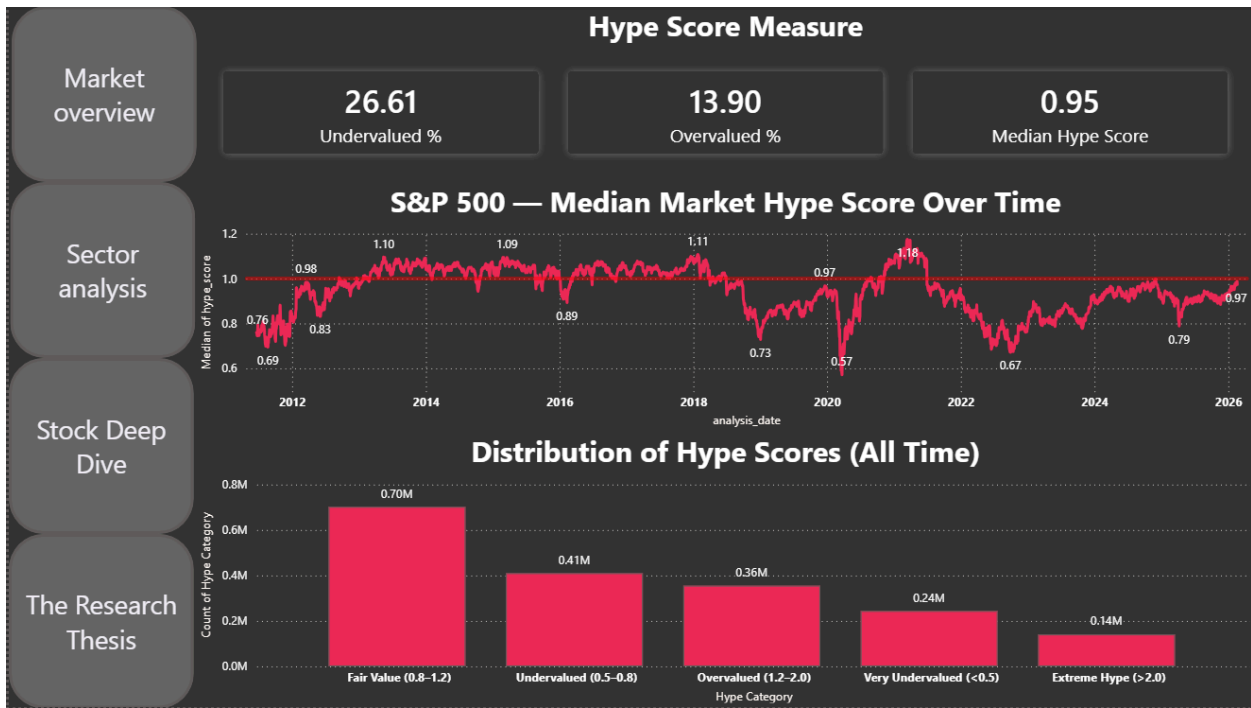
Hype Score = Current P/E ÷ 5-Year Average P/E

Survivorship Bias Treatment

The analysis includes **all historical constituents** sourced from Wikipedia's full membership records, ensuring that "fallen angels" and bankrupt companies are included, preventing an artificial inflation of historical returns.

3. Key Findings

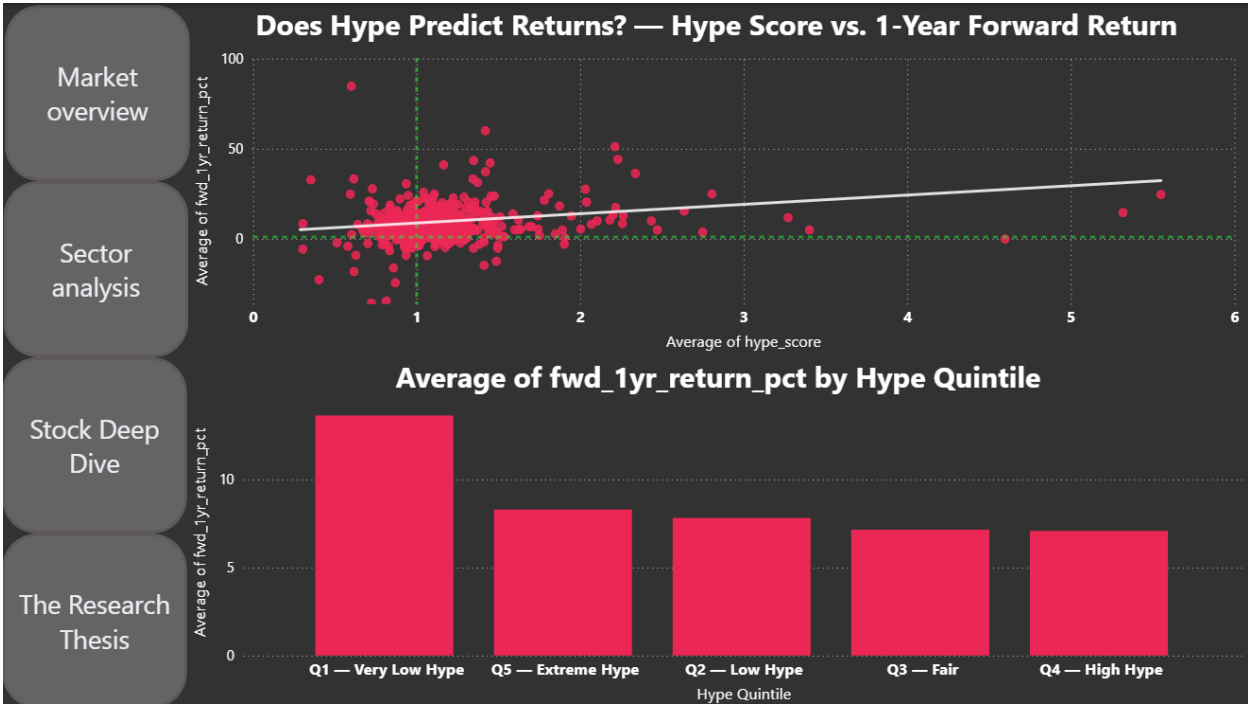
3.1 Market-Level: Hype Score Tracks Known Market Cycles



Period	Median Hype Score	Context
Early 2012	0.69–0.76	Post-2008 recovery — market still cheap

Period	Median Hype Score	Context
2013–2014	~1.10	Bull market — mild overvaluation
March 2020	0.57	COVID crash — deepest undervaluation in dataset
Mid-2021	1.16	Post-COVID speculative peak
Late 2022	0.67	Fed rate hike bear market
2025–2026	~0.97	Normalization

3.2 Value Investing Works — But the Full Picture Is More Nuanced



Quintile	Avg. 1-Year Forward Return
Q1 — Very Low Hype	~15%
Q2 — Low Hype	~8%
Q3 — Fair Value	~7.5%
Q4 — High Hype	~7.5%

Quintile	Avg. 1-Year Forward Return
Q5 — Extreme Hype	~8.5%

Deep-value stocks (Q1) generate approximately double the return of all other groups. However, the relationship is **U-shaped**: extreme hype stocks (Q5) often outperform the "boring middle" (Q2-Q4) due to high-growth outliers.

3.3 Case Study: Tesla (TSLA)

Tesla illustrates the model's self-correcting nature. During its hyper-growth phase, the Hype Score was extremely high. As profitability stabilized from 2022 onwards, the model normalized, though it correctly flagged a new overvaluation spike in late 2025.

3.4 Case Study: Amazon (AMZN)

For years, Amazon's choice to reinvest all profits resulted in near-zero EPS, making it appear chronically overvalued by this model. This highlights a limitation: the model is calibrated for mature, earnings-positive firms rather than high-reinvestment growth platforms.

3.5 Sector Patterns

4. Limitations & Caveats

- **Single-Factor Model:** Relies solely on P/E.
 - **Growth Stock Bias:** Penalizes companies with structurally low earnings due to reinvestment.
 - **Historical Data:** SEC digital data only becomes reliable from ~2009.
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5. Technical Implementation

Built as a full Python-to-PostgreSQL pipeline:

- **Storage:** PostgreSQL 15 (1.85M analytical records).
- **ETL:** Python 3.11 using `pandas` and `psycopg2`.
- **Visualization:** Power BI Desktop.

6. Summary

1. **Deep value works:** Hype Score < 0.6 doubled forward returns.
2. **U-Shaped relationship:** Both extreme value and extreme growth can outperform the middle.
3. **Risk Barometer:** Market-wide scores above 1.1 reliably signal elevated macro risk.